

Sonification

Programming and Music 2
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Sonification

Sonification focuses on the **auditory display** of data sets.

It attempts to provide a perceptual framework for dealing with abstract, non-representational (non visual) **data**.

Many creative approaches can be considered for exploring which mapping techniques are suitable for certain types of data streams. Some possibilities:

- Data-driven composition
- Audification
- Non-standard synthesis
- Unique data representations
- Data bending

Definitions

"A mapping of numerically represented relations in some domain under study to relations in an acoustic domain for the purposes of interpreting, understanding, or communicating relations in the domain under study."

(Carla Scaletti)

"The use of non-speech audio to convey information. More specifically, sonification is the transformation of data relations into perceived relations in an acoustic signal for the purposes of facilitating communication or interpretation."

(Kramer et al.)

"Aural alternatives to pie charts, line graphs and spreadsheets"

(Nicolas Collins)

Sonification

Sonification refers to mappings of data to musical parameters where **perception of the original data** is valued.

Data bending refers to mappings of data to musical parameters where the **musical output** is most valued.

Mapping is the conversion of one form or range of data to another.

Most mappings are one to one, and may involve scaling. Some mappings may be dynamic or procedural.

"Mappings are often more important than what is being mapped."

(Christopher Ariza)

Sonification

“A field in which sound articulates process is scientific sonification and auditory display . These have practical goals: mapping data onto sound, monitoring sensors, or providing auditory interfaces for computers and virtual reality systems”

(ICAD 2010)

“Stockhausen (1989) pointed out the difference between a scientific sonification and music composition. Of course, [the demonstration of a sound splitting into six parts in Kontakte] could be done more or less intelligently. . . . A physics professor would just have gone down six and a half octaves, and leave it at that. Someone else might just, well, vary it a little, make it a bit more inventive. If the same process is composed by different people and one is more imaginative than the other, then that’s all there is to say about the process, and about the difference between a physics professor and a composer in this context. Sonification is aligned with the current situation in the media arts. Today artists sonify data from diverse sources, blurring the lines of artistic and scientific projections”

(Curtis Roads, 2015).

Data Properties

Data Properties

What is usually interesting about specific data sets is discovering the possible relationships between their constituents; some of these relations may be already established, whereas others may not yet be evident.

Perceptualization is the systematic attempt to represent such relationships in data such that relationships between the constituents of the sensory rendering emerge in perception.

Data Types

Information can be broadly classified as **quantitative** (numerical) or **qualitative** (verbal).

Nominal data *classify* or *categorise*. The group and membership is a key here and examples are gender, nationality, language, genre, style etc.

Ordinal data describe the degree of order difference between different items. For example order (1st, 2nd), ranking etc.

Ratio and **Interval** data measure the relative difference between items. Examples include pitch scales, deviations, mass etc.

Data Types

Barrass (2005) has suggested **nominal/categorical** data types (e.g., different cities) could be represented by categorically changing acoustic variables, such as timbre.

Interval data may be represented by more continuous acoustic variables, such as pitch or loudness.

Sonification

The nature of a data set can takes part in possible mappings. For example, data that are **generally slow-changing**, with relatively **few inflection points** (rainfall or temperature) might be best represented with a different type of mapping than data that are **rapidly-changing with many direction changes** (EEG or stock market activity).

Some patterns for choosing the specific sound dimension of a mapping are more common. For example **pitch** is generally considered good for representing **temperature**, whereas **tempo** is not as effective.

Design

Deciding which sound events of a sonification happen **close together in time** is a fundamental design decision: temporal proximity is a strong cue for perceptual grouping.

In practice, time spent exploring design alternatives is well spent, and helps by clarifying which (seemingly natural) implicit decisions are being taken as a design evolves.

Design

The Sonification Design Space Map (SDSM) As the sonification designs become more and more sophisticated, one repeatedly answers three questions:

1. **How many data points** are likely necessary for patterns to emerge perceptually?
2. **How many and which data properties** should be represented in the design?
3. **How many parallel streams** should the design consist of?

Representations

Three important main data representation categories are:

- **Discrete**
- **Continuous**
- **Interactive**

Representations

Discrete

Discrete data representations (DDRs) are representations in which every data point (datum) is sonified with an individual auditory event.

Representations

Continuous

Continuous data representations (CDRs) treat data as analogically continuous. They rely on two preconditions: an equally spaced metric in at least one dimension and sufficient data to afford a high enough sampling rate for aural interpolation between data points. Most commonly, CDRs are used for exploring data to learn more about the system that produced it.

Representations

Interactive

In the form of graphs or maps. The term auditory graph is used in a variety of ways, often simply meaning the output of a multivariate data sonification.

Its function is to provide a sonified interface to a discrete (buffered) data set so that the relationships between the data points can be interactively investigated asynchronously.

Approaches

Alberto de Campo (2006) proposed a categorisation of sonification approaches:

(1) **Event-based**

Event-based approaches are usually concerned with a passive mode of interaction. The mapping is created per event such that certain events cause certain sounds to be played. A simple example are alerts or notifications. More complex event-based approaches usually focus on mapping or change of behaviour in time but with a relatively simplistic causality model.

Approaches

(2) **Model-based**

In Model-based approaches the designer usually creates a artificial model with which the listener interacts such that the model's "properties are informed by the data" (de Campo, 2006, p. 2).

Often the user's input drives the sonification such that the model is "*a dynamic system capable of a dynamic behavior that can be perceived as sound*" (**Bovermann, Hermann, & Ritter, 2006, p. 78**).

The structure of a data set is exposed to the user during the time he interacts with it and therefore the design requires external input. These approaches of allow for quite complex data sets.

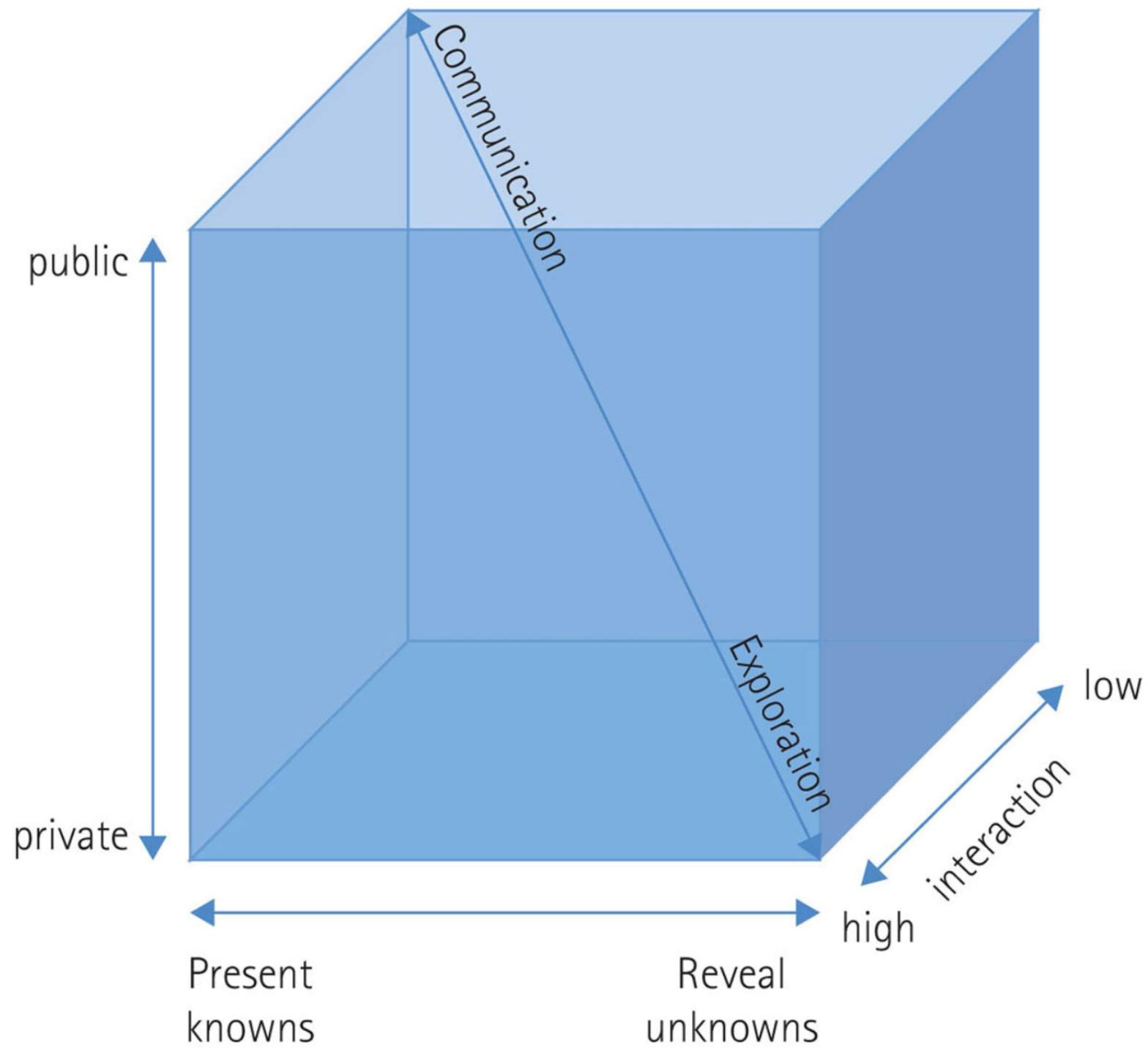
Approaches

(3) **Continuous.**

Continuous sonification deal with time series and data sampled at a rate such that a data trajectories can be directly translated into sound.

Audification is the most known method of continuous sonification, where waveforms of periodic data are directly translated into sound.

The Sonification Space



(Scaletti)

Sonification Process (Scaletti)

Process: Sonification is taking data generated by a model, captured in an experiment, or otherwise gathered through observation, and mapping those data to one or more parameters of an audio signal or sound synthesis model.

Goal: The goal of sonification is to better understand, communicate, or reason about the original model, experiment, or system.

Loop-back path: Typically, any new understanding obtained through sonification generates new questions, which can loop back to the beginning of the process again, this time with a change or refinement to the mapping or perhaps even with a new set of data, a new set of experiments, or modifications to the mathematical model that generated the data.

The Mapping Problem

The Mapping Problem

An idea in one **domain** to be manifested in another one.

The term indicates that there is a problem (such as a math problem) and that there is no obvious general solution to it. Since in most cases we are dealing with artistic practice, the problem is not likely to have a well-defined solution.

Parametric Mapping

Parameter mapping is the most widely used sonification technique for representing high-dimensional data as sound. Typically, data dimensions are mapped to sound parameters, which can be:

- **physical** (frequency, amplitude)
- **psychophysical** (pitch, loudness)
- **perceptually coherent complexes** (timbre, rhythm).

Many data dimensions can be listened to simultaneously. Mapping can be flexible, and the mappings can be easily changed, allowing different aural perspectives of the same data.

Parametric Mapping

In practice parameter mapping requires a working knowledge of how the parameters interact with each other perceptually.

Linear changes in one domain can produce nonlinear effects in another one.

An example could be how to map **physical space** to an **auditory** one.

Perhaps space could be transferred to a temporal view?

Parametric Mapping

“Auditory time involves concepts like repetition, recapitulation, backward and forward reference, morphological similarity. ”

(Larry Polansky).

How can these be derived from data or have visual similarities?

Mapping Considerations

Complex mappings may be **categorical** (feature-based) where insights into the data and how to classify it might inspire the mapping.

Simplifying data for is also common, to find out how much information is needed to cause effect. For example *'I am going to class'* is less complex than *'Im a going to class at Sonolgy from 13:00 - 15:00'*.

Context refers to the addition of additional displays to provide reference. A simple example would the addition of a specific sound for a specific city from where its population or temperature would drive a sonification.

Polarity & Scaling

Mappings often require a choice of **polarity**.

An example might be where **pitch** should **increase** to represent higher temperatures (a positive mapping polarity). Another example might be pitch that is **decreasing** to represent increasing space (a negative polarity).

Scale determines by how much a change in data (for example temperature) might affect change in sound (for example pitch).

Scaling (and the understanding of a domain) is critical if a sonification is intended to be accurate instead of simple trend estimations

Practice

Composition

Computer-based composition tools exist for exploring algorithmically generated structures at both micro(sound (sound synthesis) and macro (gestural) compositional levels.

Some approaches ('free data' or 'glitch') involve a kind of **data music** that uses arbitrarily formatted digital documents as control data for sound synthesis.

The outcome of such procedures is often unpredictable but could be influenced by a belief (possibly arbitrary) in the outcome of a certain sonification.

Data Bending

Data bending differs from 'data sonification' where more emphasis is on treating directly the sonic signal of a data set instead of attempting to map in a meaningful way its properties.

The approach might also prefer 'misinterpretation' of data and the surprises/novelty it might bring forth.

The aesthetics of failure involve a certain technological critique, a post-digital attitude that attempts to bring forth the imperfection of computer music and its tools.

Aesthetics of Failure

“Computers have become the primary tools for creating and performing electronic music, while the Internet has become a logical new distribution medium. For the first time in history, creative output and the means of its distribution have been inextricably linked. [...]” The glitch movement can be seen as a natural progression in electronic music. In this new music, the tools themselves have become the instruments, and the resulting sound is born of their use in ways unintended by their designers. Commonly referred to as sound ‘mangling’ or ‘crunching,’ composers are now able to view music on a microscopic level. [...] Glitch artists tends to be based on experimentation rather than empirical investigation. In this fashion, unintended usage has become the second permission granted.”

(Kim Cascone, 2002)

Data music

At a representational end is data music that uses data for decision making, development in time and to explore compositional ideas.

Algorithmic Composition (or computer-aided composition) can also relate with sonification through concepts of experimental music *"a music in which some more or less unknown result emanates from some experimental "hypotheses"*. **(Larry Polansky)**

However these also differ considerably due to their intentions.

"Characteristics and qualitates that are essential to reveal information used in sonification such as "clarity", "efficiency" and "economy" may be only of secondary or of no interest within a aesthetic context" **(Larry Polansky)**.

Data music

"Use of streams of data for compositional and sound generating purposes is very familiar to electro-acoustic composers, artists and performers. Sonication in the artistic view is pushing the idea a little farther—instead of algorithms, real world data are used, although reaching their exact, understandable representation becomes additional or even insignificant, rather than essential."

(Jonas Gruska)

Audification



<http://michaelchinen.com/fuckingwebbrowser/>

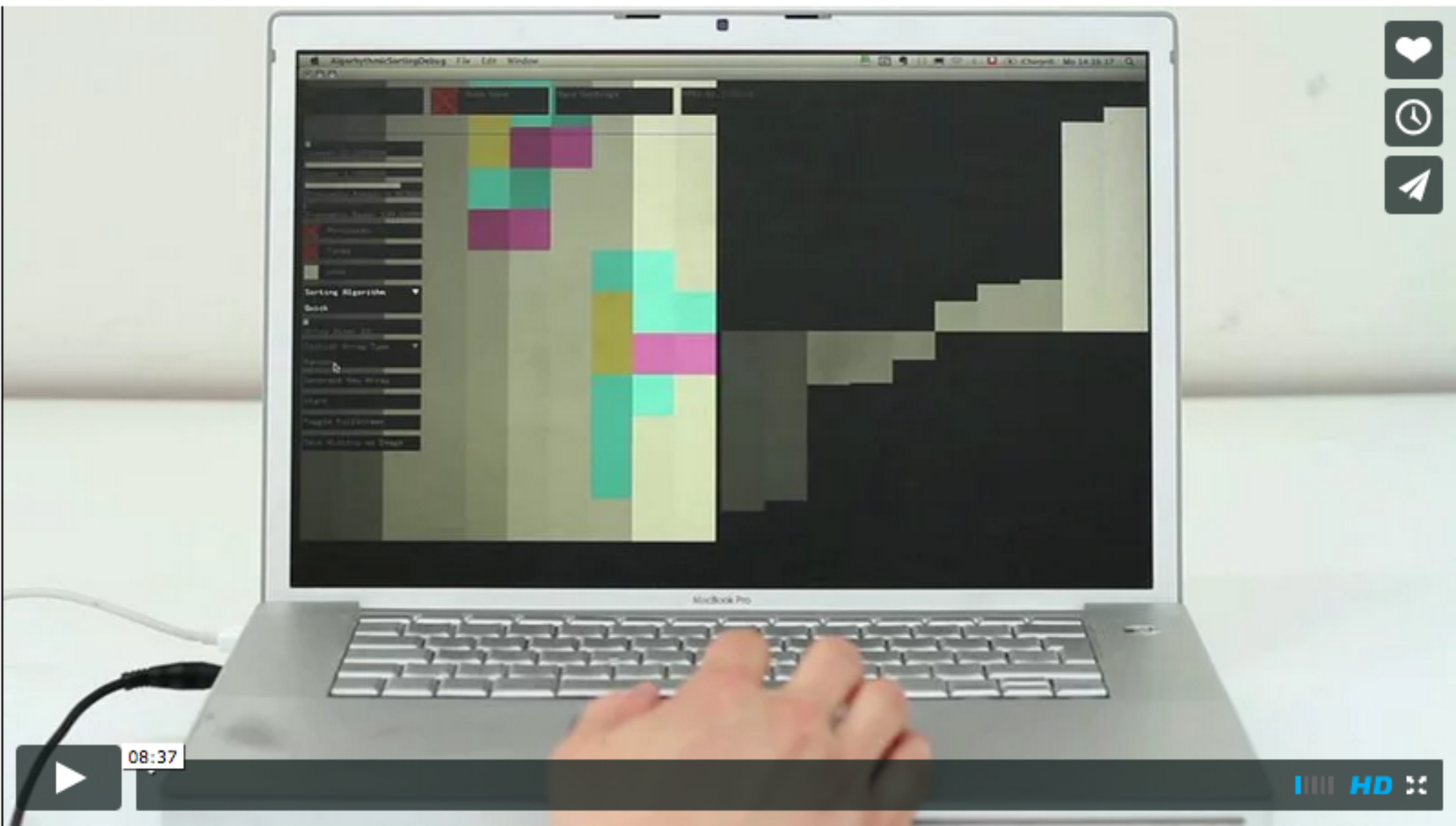
Audification

"Sonification involves the study of relations in a data set and their transformation into sonic relations that may be perceived by a listener. The framework of these sound relations is often intended to be musical, or derived primarily from musical ideas."

(Chinen as quoted by Casper Shipper, 2011)

"FuckingWebBrowser is a simple open-source WebKit based browser (for which there are hundreds of tutorials including many of annoying ones on youtube how to make in less than 2 minutes on a mac.) I added sonification, which converts the memory state into audio. The quality of sound is noise, but an overall structure is recognizable due to changes in the user interface (while the images load, while I mouse over/scroll, etc), which directly affect computer memory."

Rhythms of Sorting



08:37

HD

 **AlgoRhythmic Sorting | Sorting Algorithms**
from Shintaro Miyazaki 3 years ago / CC BY-SA NOT YET RATED

Politiken der Frequenz

Marcus Schmickler & Julian Rohrerhuber (eMEGO 191)

"Following an hypothesis according to which "Music and Economics share a fundamental object: number", Schmickler and Rohrerhuber's project POLITIKEN DER FREQUENZ circles around the acoustic rendering of number concepts. Inspired by Alain Badiou's Le Nombre et les Nombres and accompanied by mixed choir, the piece attempts to question the apparent immediacy of number that allows calculation to govern today's economy, social sciences and everyday life. [...] A number and its properties can be considered either as an outcome of a dynamic process, like measurement or counting, or (and) as something that exists independently, and is a precondition necessary for such operations in the first place. [...] drawing from the history of mathematics, a sonification of number brings to light something fundamentally different, namely its acoustic shimmer, its glint."

Marcus Schmickler & Julian Rohrhuber
Spectral Digits (2014)



Auditory Sieve

“The work of Auditory Sieve exercises the objectual logics of computational listening - wavelet analysis and re-synthesis - in conjunction with the sieve algorithm as introduced by Iannis Xenakis. While the technique of wavelet analysis and re-synthesis can be generalised as an “intimate transformation” [Risset and Wessel, 1982] sensitive to the temporal and spectral intricacies of the input sound material, sieves impose a formal template from outside with no regard to what consists its object². In a way, wavelets and sieves represent two opposing polarities, dialectical tropes of bottom-up and top-down strategies. The work of ‘Auditory Sieve’ attempts to mobilise a process of creative grafting across and between these two strategies and their temporal resolutions.”

Marcin Pietruszewski - Sieve Decantation (i) (2020)



```

(
  NF(\iop, {|freq=78, mul=1.0, add=0.0|
    var noise = LFNoise1.ar(0.001).range(freq, freq + (freq * 0.1));
    var osc = SinOsc.ar([noise, noise * 1.04, noise * 1.02, noise * 1.08],0,0.2);
    var out = DFm1.ar(osc,freq*4,SinOsc.kr(0.01).range(0.92,1.05),1,0,0.005,0.7);
    HPF.ar(out, 40)
  }).play;
)

(
  NF(\dsc, {|freq = 1080|
    HPF.ar(
      BBandStop.ar(Saw.ar(LFNoise1.ar([19,12]).range(freq,freq*2), 0.2).excess(
        SinOsc.ar( [freq + 6, freq + 4, freq + 2, freq + 8])),
        LFNoise1.ar([12,14,10]).range(100,900),
        SinOsc.ar(20).range(9,11)
      ), 80)
    ).play;
)

var <>pindex, <>cindex;

initialize {
  if(pindex.isNil, { pindex = 1000 });
  if(cindex.isNil, { cindex = 2000 });
}

clearProcessSlots {
  pindex = 1000;
  (this.pindex - 1000).do{|i| this[this.pindex+i] = nil; }
}

clearOrInit {|clear=true|
  if(clear == true, { this.clearProcessSlots() }, { this.initialize() });
}

transform {|process, index|
  if(index.isNil && pindex.isNil, {
    this.initialize();
  });

  pindex = pindex + 1;
  this[pindex] = \filter -> process;
}

control {|process, index|
  var i = index;

  if(i.isNil, {
    this.initialize();
    cindex = cindex + 1;
    i = cindex;
  });

  this[i] = \pset -> process;
}

(
  NF(\depfm, {|freqMin=5, freqMax=20, mul=20, add=80, rate=0.5, modFreq=2100, index=0.3, amp=0.2|
    var trig, seq, freq;
    trig = Dust.kr(rate);
    seq = Diwhite(freqMin, freqMax, inf).midicps;
    freq = Demand.kr(trig, 0, seq);
    HPF.ar(PMOsc.ar(LFCub.kr([freq, freq/2, freq/3, freq/4], 0, mul, add),
      LFNoise1.ar(0.3).range(modFreq,modFreq*2), index) * amp, 50)
  }).play;
)

```

Exercises

Exercises

1. Implement a simple sonification strategy for sonifying the Sonology course data.
2. Implement a simple sonification strategy for sonifying the senator and votes data.
3. Implement a simple sonification strategy for sonifying the earthquake data.